

axionbloch: an Open-Source Python Package for Simulating Axion-Induced Spin Dynamics

Yuzhe Zhang ^{1,2,3} 

¹ Johannes Gutenberg-Universität Mainz, 55099 Mainz, Germany

² Helmholtz Institute Mainz, 55128 Mainz, Germany

³ GSI Helmholtzzentrum für Schwerionenforschung GmbH, 64291 Darmstadt, Germany

Abstract

The interaction of ultralight bosonic dark matter with spins can be interpreted as a pseudo-magnetic field acting on normal matter. Such interactions can be modeled as usual magnetic interactions using spin-evolution (Bloch) equations. *axionbloch*, an open-source Python package for simulating spin dynamics induced by both usual and exotic interactions, is presented. The numerical simulations serve as a tool for deriving axion signal signatures, which are crucial for designing experimental searches and data analysis. Simulations are calibrated against theoretical expectations, ensuring the accuracy of the simulated signals. *axionbloch* is available at <https://github.com/Yuzhe98/AxionBloch>, allowing researchers to simulate pseudomagnetic signals under specific configurations of the axion models and experimental setups. The package is documented at <http://axionbloch.readthedocs.io/> and includes example scripts for application.

Keywords: axion dark matter; nuclear magnetic resonance; Bloch equations; spin dynamics; Python package

1. Introduction

The axion was originally proposed to resolve the strong *CP* problem in quantum chromodynamics [1–4]. Here *C* denotes charge conjugation and *P* denotes parity. Later, more general axionlike particles (ALPs) were proposed in various extensions of the Standard Model associated with spontaneously broken global symmetries [5–8]. Concerning the cosmological implications, axions and ALPs are also well-motivated dark matter candidates [9–12]. In this paper, axions and ALPs are referred to as “axions” for simplicity.

Axions may possess couplings to ordinary Standard Model particles [13], including photons, gluons, and fermions. Crucially, axion field gradients (∇a) may couple to fermionic spins through a pseudoscalar interaction. The Hamiltonian for this interaction can be expressed as

$$\mathcal{H}_{\text{int}} = -g_{\text{aNN}} \nabla a \cdot \mathbf{S}, \quad (1)$$

where g_{aNN} is the axion-spin coupling strength, and $\mathbf{S}$ is the spin operator. The axion field gradient can be regarded as a pseudomagnetic field that mimics the behavior of a real magnetic field, where the Hamiltonian can be written as

$$\mathcal{H} = -\hbar \gamma \mathbf{B} \cdot \mathbf{S}, \quad (2)$$

where γ is the gyromagnetic ratio of the spin, and $\mathbf{B}$ is the magnetic field. The pseudomagnetic field induces spin dynamics, providing a pathway for detection in precision measure-

Received:

Accepted:

Published:

Copyright: © 2026 by the author.

Submitted to *Universe* for possible open access publication under the terms and conditions of the [Creative Commons Attribution \(CC BY\)](https://creativecommons.org/licenses/by/4.0/) license.

ment experiments such as nuclear magnetic resonance (NMR) and comagnetometer-based searches [14–37]. There have been dedicated studies of the axion-induced spin dynamics [38–40], providing axion signal signatures and experimental sensitivity to axion fields. However, to our knowledge, there have not been studies taking advantage of numerical methods to verify the derived axion signal signatures. On the other hand, as more axion models and experiments are being proposed, such as cosmic axion background [41,42], fine-grained axion [43], and quadratically coupled axion [44,45], a tool is needed to systematically simulate the axion-induced spin dynamics under various physical conditions and axion field configurations.

In this work, *axionbloch*, an open-source package designed to simulate spin dynamics induced by axion fields, is introduced. The interface is implemented in Python, a programming language known for its simplicity and readability, while the underlying numerical integration is implemented in C++ for efficiency. The package provides a flexible framework for modeling axion fields and experimental configurations. By numerically solving spin-evolution equations, *axionbloch* enables the prediction and analysis of experimentally observable signals.

2. Capabilities

In this section, the main components of *axionbloch* are described. The section starts with the units, physical quantities, and constants (Section 2.1). Then, it continues with the physics and the algorithm for the numerical simulation (Section 2.2), after which the calibration for the simulation follows (Section 2.3). In the end, an example simulation of axion-induced signals is provided (Section 2.4).

2.1. Units, physical quantities, and constants

The package takes advantage of *astropy*¹ [46] to deal with units and physical quantities. Using *astropy.units*, physical quantities are created by attaching units to Python scalars or *NumPy*² [47] arrays. Quantities can be converted between dimensionally equivalent units using the `to()` method. In *astropy*, unit definitions are included in both the International System of Units (SI) and the Centimeter-Gram-Second (CGS) systems. Values in SI or CGS units can be found in the quantity’s properties `.si` and `.cgs`. A code example can be found in Figure 1.

In *axionbloch.constants*, a few commonly-used constants in axion and NMR research, such as gyromagnetic ratios, are defined. The constants are instances of *astropy.units.Quantity*; therefore, all operations on physical quantities are supported for the constants.

2.2. Algorithm

2.2.1. Spin dynamics

Considering the current experiments looking for axion-induced spin dynamics, *axionbloch* focuses on spin-1/2 magnetic resonance. Different spins may be addressed in future updates. Magnetization is used as the observable to characterize the spin en-

¹ <https://astropy.org/>

² <https://numpy.org/>

```

Define a quantity from scalar and unit
>>> from astropy import units as unit
>>> B = 1.0 * unit.gauss
>>> B
<Quantity 1. G>
>>> B.si
<Quantity 0.0001 T>

>>> import numpy as np
Import constants and use them with scalars
>>> from axionbloch.constants import gamma_p
>>> from axionbloch.dependency import PI
Find the 90 degree pulse duration
>>> t90 = PI / 2 / (gamma_p * B)
>>> t90.to(unit.microsecond)
<Quantity 58.71648793 us>

Operation on an array of quantities with numpy
>>> tStamps = np.array([0, 1 / 3, 1]) * t90
>>> phases = gamma_p * B * tStamps
>>> np.sin(phases)
<Quantity [0. , 0.5, 1. ]>

```

Figure 1. Examples of operations on physical quantities using `astropy.units` and `axionbloch.constants`.

semble. In the presence of magnetic fields or axion-induced pseudomagnetic fields, the magnetization $\mathbf{M}$ evolves as

$$\frac{dM_x}{dt} = \gamma(\mathbf{M} \times \mathbf{B}_{\text{eff}})_x - \frac{M_x}{T_2}, \quad (3)$$

$$\frac{dM_y}{dt} = \gamma(\mathbf{M} \times \mathbf{B}_{\text{eff}})_y - \frac{M_y}{T_2}, \quad (4)$$

$$\frac{dM_z}{dt} = \gamma(\mathbf{M} \times \mathbf{B}_{\text{eff}})_z - \frac{M_z - M_0}{T_1}, \quad (5)$$

where γ is the gyromagnetic ratio, $\mathbf{B}_{\text{eff}}$ is the total effective field, T_1 and T_2 are relaxation times, and the subscripts x, y, z denote the Cartesian components. The bias field is usually much stronger than the pseudomagnetic field. Therefore, the rotating coordinate frame (RCF) is adopted to remove the Larmor precession due to the bias field, allowing efficient simulation of the slower axion-induced dynamics of interest.

The transverse relaxation time, characterized by time constant T_2^* , includes the T_2 relaxation and the dephasing due to the inhomogeneity of the bias field. Such inhomogeneity of the bias field is simulated by sampling the spread of the bias fields in the `Magnet` instance, and solving the Bloch equations for individual bias fields.

2.2.2. Numerical integration

To derive the time evolution of the magnetization, `axionbloch` numerically integrates the Bloch equations using the fourth-order Runge–Kutta (RK4) method. Given a time step Δt , an effective magnetic field, and initial magnetization $\mathbf{M}^0$, the magnetization at the $(n + 1)$ -th step can be found via

$$\mathbf{M}^{n+1} = \mathbf{M}^n + \frac{\Delta t}{6} (\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4). \quad (6)$$

Here

$$\mathbf{k}_1 = \left. \frac{d\mathbf{M}}{dt} \right|_{\mathbf{M}=\mathbf{M}^n}, \quad (7)$$

$$\mathbf{k}_2 = \left. \frac{d\mathbf{M}}{dt} \right|_{\mathbf{M}=\mathbf{M}^n + \frac{1}{2}\mathbf{k}_1\Delta t}, \quad (8)$$

$$\mathbf{k}_3 = \left. \frac{d\mathbf{M}}{dt} \right|_{\mathbf{M}=\mathbf{M}^n + \frac{1}{2}\mathbf{k}_2\Delta t}, \quad (9)$$

$$\mathbf{k}_4 = \left. \frac{d\mathbf{M}}{dt} \right|_{\mathbf{M}=\mathbf{M}^n + \mathbf{k}_3\Delta t}, \quad (10)$$

where $d\mathbf{M}/dt$ can be found using the Bloch equations [Eqs. (3–5)]. This fourth-order method yields a local truncation error of $\mathcal{O}(\Delta t^5)$, while the accumulated error over $N = 1/\Delta t$ steps is of order $\mathcal{O}(\Delta t^4)$. The time step Δt is usually set to be at least one order of magnitude smaller than the characteristic timescales of the system, such as relaxation times.

2.2.3. Implementation

Using object-oriented programming, the algorithm is implemented in a modular way to separate experimental configuration, axion field modeling, and numerical integration, allowing for flexible configuration of simulations. A diagram of the architecture of axionbloch is shown in Figure 2.

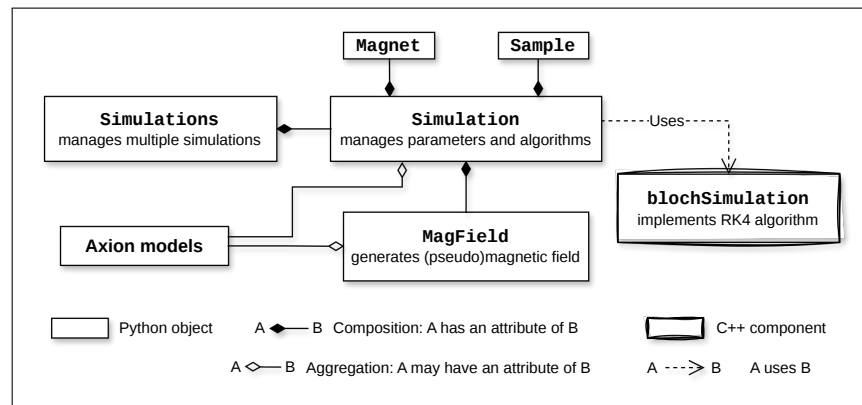


Figure 2. Architecture of axionbloch. Each *Simulation* instance imports parameters from *Magnet*, *Sample*, and *MagField* instances. For axion simulations, axion-model objects are required both for *Simulation* to configure the simulation and for *MagField* to generate the pseudomagnetic field. *Simulations* manages one or multiple *Simulation* instances. Numerical integration is performed via the *blochSimulation* API, which provides Python access to the underlying C++ implementation.

On the Python side, classes *Sample*, *Magnet*, and *MagField* encapsulate the relevant experimental parameters, including sample relaxation times, magnetic field strength, and inhomogeneity. The axion models are implemented by dedicated classes, such as *MilkyWayAxionHalo*, which carry the properties of the axion fields. Together with the method *MagField.setAxionFields()*, the pseudomagnetic fields can be generated. Different axion models can be incorporated as separate classes without modifying the simulation framework. *Simulation* is handled within the *axionbloch.SimuTools* module. In particular, the classes *Simulation* and *Simulations* manage the interaction between experimental configurations and axion models, organizing the parameters and execution of single or multiple simulation runs.

The RK4 algorithm is implemented in a dedicated C++ library. The application programming interface (API) for this library is *blochSimulation*. This C++ backend is

interfaced with Python using pybind11,³ enabling cooperation between high-level Python workflow management and low-level efficient numerical computation.

2.3. Calibration

Accurate calibration is essential to ensure that the simulated signals reliably reflect the underlying physical system and can be compared with experimental observables. In particular, calibration examines the correspondence between the NMR signal and model parameters, including magnetic fields, nuclear spin gyromagnetic ratio γ , and relaxation times (T_1 , T_2 , and T_2^*). It allows one to verify the correctness and stability of the numerical implementation. In this section, examples of calibration simulations are presented along with theoretical predictions, demonstrating the methodology of calibration. A quantitative analysis is included at the end of the section. For the examples in the section, an ethanol sample and an axial bias field $\mathbf{B}_0 = B_0 \hat{\mathbf{e}}_z$ are used.

2.3.1. Pulsed-NMR: free decay and spin echo

In the pulsed-NMR calibration, one 90° and one 180° magnetic-field pulse are applied in the simulation. The magnetization is expected to [48,49]:

- oscillate at the Larmor frequency;
- be tipped by 90° to the x-y plane by the 90° pulse;
- free decay with relaxation time T_2^* in the absence of pulses;
- refocus (echo) after 180° pulses;
- show decaying amplitude of echoes with relaxation time T_2 .

The behaviors described above can be found in the calibration example illustrated in Figure 3 (a), indicating correct simulation behavior. With such a benchmark, magnetic pulses, Larmor precession, and transverse relaxations (T_2 and T_2^*) can be calibrated.

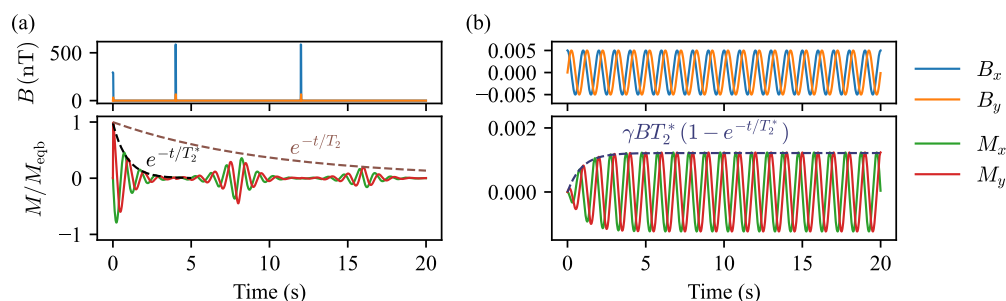


Figure 3. Simulated magnetic fields in x and y directions (upper row) and induced NMR signals (lower row). The magnetizations plotted are normalized by equilibrium magnetization. (a) After 90° and 180° magnetic-field pulses in x-y plane, free decay (relaxation time T_2^*) and echo signals (relaxation time T_2) are generated. (b) Excited by a weak CW magnetic field, the NMR signals increase with $\gamma B T_2^* (1 - e^{-t/T_2^*})$, where $B = \sqrt{B_x^2 + B_y^2}$ is the magnitude of the excitation.

2.3.2. Continuous-wave NMR

In the continuous-wave (CW) NMR calibration, an oscillating magnetic field is applied. When the magnetic field strength B is weak ($\gamma B T_2^* \ll 1$) and on resonance, the transverse magnetization grows with $\gamma B T_2^* (1 - e^{-t/T_2^*})$ [49]. One such example is illustrated in Figure 3 (b). The simulation produces the expected results. With such a benchmark, the Larmor precession and T_2^* relaxation can be calibrated.

³ <http://pybind11.readthedocs.io/>

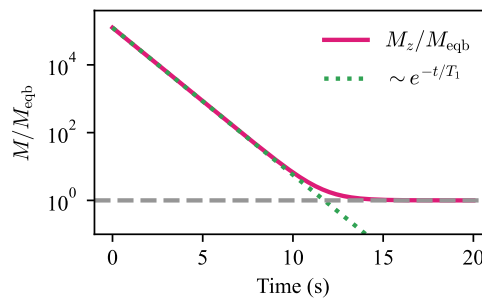


Figure 4. Simulated T_1 relaxation of a hyperpolarized sample in a static bias field. The axial magnetization M_z is normalized by the equilibrium magnetization M_{eqb} .

2.3.3. Hyperpolarized sample

To achieve better sensitivity to the axion field, hyperpolarization is favored in experiments. Compared to thermal polarization at room temperature, hyperpolarization techniques can boost the polarization by orders of magnitude to the order of unity. The *axionbloch* package handles the polarization by the *Sample* class. A 1 %-polarization ethanol sample is used as an example demonstrating the evolution of polarization due to the T_1 relaxation. At first ($t \ll T_1$), the polarization should exponentially decay with time constant T_1 , until it reaches equilibrium after many T_1 . As can be seen in Figure 4, the simulated T_1 relaxation agrees with the expected behavior. More of such simulations can be applied in calibrating T_1 relaxation.

2.3.4. Quantitative evaluation

The accuracy of simulations is evaluated by the discrepancy between the theoretically expected magnetization and simulation results, characterized by

$$\chi^2 = \frac{\sum |M_{\text{expected}} - M_{\text{simu}}|^2}{\sum |M_{\text{simu}}|^2}. \quad (11)$$

Here M_{expected} and M_{simu} are magnetization expected and simulated, respectively. $\sum$ denotes the sum over integration steps. A well-calibrated simulation should give $\chi^2 \ll 1$. Calibration scripts are stored in the test suite of *axionbloch*. In each calibration, parameters such as Larmor frequency and relaxation times are varied over orders of magnitude to cover all possible cases in simulation. The results of the calibrations are summarized in Table 1. The discrepancies are generally small ($\chi^2 < 10^{-4}$). If a certain simulation precision is required, the χ^2 value serves as a quantitative reference.

Table 1. Summary of calibrations. χ^2 indicating the discrepancy is defined in Equation (11); $\chi^2 \ll 1$ indicates a well-calibrated simulation. Here ν_L stands for Larmor frequency, determined by the bias magnetic field. ν_{RCF} is RCF frequency. ν_L and ν_{RCF} are varied from 1 kHz to 1 GHz. The bias field inhomogeneity is swept from 0.1 ppm to 20 ppm. T_1 ranges from 1 ms to 1 ks. The integration step size is chosen to be $T_2^*/500$. Finer step sizes yield lower χ^2 at increased computational cost. More technical details of the calibrations can be found in the test suite.

Calibration	Parameters calibrated	Varied parameters	χ^2	Dominant source of error
Pulsed NMR: free decay	Pulse amplitude, ν_L , T_2^* , T_2	ν_{RCF}	$< 10^{-4}$	Finite number of steps for hard pulses.
CW-NMR	ν_L and T_2^*	ν_{RCF}	$< 10^{-5}$	Finite integration step size.
T_1 relaxation	T_1	ν_{RCF} , ν_L , initial polarization	$< 10^{-7}$	Finite integration step size.

2.4. Axion simulation

Similar to the calibration examples, axion-induced NMR signals are generated by solving the Bloch equations with the axion-induced pseudomagnetic field as part of the effective magnetic field. An example of a commonly considered model – axion constituting the Milky Way galactic dark matter halo [50,51] is implemented in the class `MilkyWayAxionHalo`. In this model, the axion field is treated as an oscillating field with a frequency determined by the axion mass, and a linewidth (characterizing the stochasticity) determined by the velocity dispersion of the virialized dark matter halo. The sample is chosen to be ^{129}Xe , polarized to 50 % initially, and the bias field is applied along the z direction. The details of the simulation configuration can be found in the code example in Figure 5. Readers are referred to Figure 2 which illustrates the relationship between the objects in the code example. The simulated pseudomagnetic fields and the amplitude of the induced transverse magnetization are shown in Figure 6, in which the signal increases with time at the beginning of the simulation due to the pseudomagnetic field, and then decays due to the T_1 relaxation. The efficiency of the simulation is limited by time consumption for the numerical integration of the Bloch equations. The total number of integration steps N can be found by

$$N = \text{Simulation.rate} \times \text{Simulation.duration} \times \text{MagField.numFields} \times \text{Magnet.numPt}, \quad (12)$$

where `Simulation.rate` and `Simulation.duration` are the simulation rate and duration, `MagField.numFields` is the number of random pseudomagnetic fields generated to account for the stochasticity of the axion field, and `Magnet.numPt` is the number of bias fields sampled to account for the inhomogeneity of the bias field. For the example in Figure 5, the total number of integration steps is 7×10^8 , and the runtime is 2.7 s on a personal computer (CPU: Intel® Core™ Ultra 7 155H; base speed: 1.40 GHz). Such moderate runtime allows users to explore different parameter regimes with many simulations.

3. Development

Development of `axionbloch` is managed using `git`,⁴ with the code hosted on a public GitHub repository.⁵ Changes to the main codebase follow a *pull request* workflow, allowing contributions to be reviewed before being merged. Users can participate by forking the repository, making modifications in their fork, and submitting pull requests for review by the developers. As `axionbloch` is intended as a collaborative resource, user contributions and suggestions are encouraged and appreciated.

`axionbloch` includes a suite of automated tests to ensure correctness and reliability. The tests are implemented using the `pytest` framework,⁶ which allows for calibration of physical processes and testing of functions, classes, and modules. The test suite covers core functionalities such as:

- initialization and manipulation of classes;
- numerical integration of the Bloch equations;
- magnetic fields and relaxations;
- handling of each axion field configuration.

Tests can be executed locally. In the future, tools may be added for online testing considering the demand of testing.

⁴ <http://git-scm.com>

⁵ <https://github.com/Yuzhe98/AxionBloch>

⁶ <https://docs.pytest.org/>

```

# Example script to run axion NMR simulations
# numpy, matplotlib, astropy dependency
from axionbloch.Apparatus import Magnet
# Gyromagnetic ratio and magnetic dipole moment of Xe-129
from axionbloch.constants import gamma_Xe129N, mu_Xe129N
from axionbloch.dependency import *
from axionbloch.MilkyWayAxionHalo import MilkyWayAxionHalo
from axionbloch.Sample import Sample
# classes for simulations
from axionbloch.SimuTools import MagField, Simulations
from axionbloch.SimuTypes import SimuParams
# Define the Xe-129 sample with gyromagnetic ratio,
# mass density, molar mass, number of spins per molecule,
# relaxation times, volume, magnetic dipole moment,
# temperature, and polarization.
sample = Sample(
    name="Liquid Xe-129",
    gamma=gamma_Xe129N,
    massDensity=3.1 * unit.g * unit.cm ** (-3),
    molarMass=131.29 * unit.g / unit.mol,
    numOfSpinsPerMolecule=1 * unit.one,
    T2=10 * unit.minute,
    T1=15 * unit.minute,
    vol=1 * unit.cm**3,
    mu=mu_Xe129N, # magnetic dipole moment
    temp=163 * unit.K,
    pol=1 * unit.percent,
    verbose=False,
)
# Define the axion field with axion Compton frequency
# and axion-nucleon coupling strength
axion = MilkyWayAxionHalo(
    name="Milky Way Axion Halo",
    nu_a=1 * unit.kHz,
    g_aNN=1.0e-9 * unit.GeV ** (-1),
    verbose=False,
)
# Set the bias field strength, direction, and homogeneity
magnet = Magnet(
    B0=axion.nu_a_eff / (sample.gamma / (2 * PI)),
    FWHM=2 * ppm,
)
# rms amplitude of pseudomagnetic field
B_a_rms = (axion.getRabiFreq() / \
            (sample.gamma / (2 * PI))).to(unit.T)
# Bundle all inputs into one dictionary
params: SimuParams = {
    "key_info": {"nu_a": axion.nu_a},
    "axion": axion,
    "sample": sample,
    "magnet": magnet,
    "excField": MagField(),
    "B_a_rms": B_a_rms,
    # Number of random field realizations.
    "numFields": 1000,
    "rand_seed": 10, # random seed
    # amplitude, polar and azimuthal angle
    # of the initial magnetization
    "init_M": None, # use None when setting M automatically
    "init_M_theta": 0 * unit.rad,
    "init_M_phi": 0 * unit.rad,
    # sampling rate and duration of the time series
    "rate": 1 * unit.Hz,
    "duration": 4000 * unit.s,
}
# Create and execute the simulation job collection
simulations = Simulations(all_params=[params])
# run the simulation
simulations.run(verbose=True)
# Post-process results with summary stats and plotting
for i, item in enumerate(simulations.pool):
    item.simu.displayTrjries()
# Save to .pkl file for later analysis
simulations.saveToPkl(dir="path_to_save")

```

Figure 5. Example code of an axion simulation.

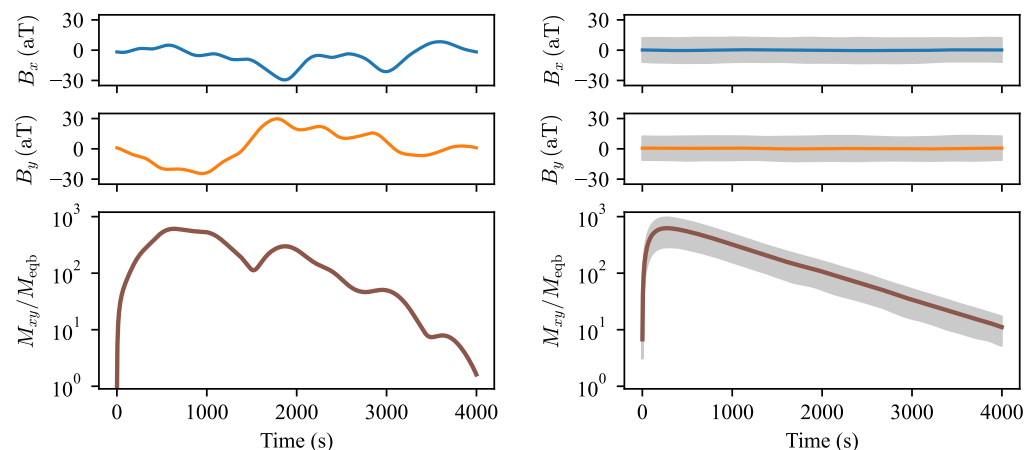


Figure 6. Simulations of local virialized axion dark matter halo. From top to bottom: pseudomagnetic fields in the transverse plane (B_x and B_y) and the transverse magnetization normalized by the equilibrium magnetization (M_{xy}/M_{eqb}). Left: one single simulation. Right: 1000 simulations to account for the stochasticity. The solid lines indicate the value (left column) or mean value (right column) of B_x , B_y , and M_{xy} , while the gray shadows indicate the $\pm\sigma$ (standard deviation) range. The sample for the simulations is assumed to be polarized to 50 % initially.

Documentation for `axionbloch` is available on Read the Docs.⁷ It is built using Sphinx⁸ and provides detailed explanations of the API, along with instructions for installation, configuration, and usage.

As numerous new axion models and experiments are emerging, the functionality of `axionbloch` may extend so that it can be a tool for updating our knowledge of the induced spin dynamics. Furthermore, capabilities for simulating zero-to-ultra-low-field (ZULF) NMR will be implemented in future versions.

4. Conclusions

`axionbloch` has been presented as a framework for simulating spin dynamics induced by magnetic fields or axion fields under a variety of physical scenarios. The package adopts SI units so that the inputs and outputs are directly comparable to experimental parameters and observables. Numerical integration of the Bloch equations is performed using a fourth-order Runge–Kutta method in the rotating coordinate frame, enabling stable and efficient simulation. Comprehensive calibrations have been performed to ensure the accuracy of the simulations. An example simulation of axion-induced signals from a local virialized axion dark matter halo is presented to demonstrate the package’s capabilities.

By providing an open-source, well-documented, and tested tool, `axionbloch` aims to facilitate the study of axion-induced spin dynamics in precision measurement experiments and to serve as a platform for further method development and incorporation of new axion models.

Note that this package can be of interest to the NMR community for its applications in, for example, educational purposes. For example, students can specify pulse sequences for the simulation and see the visualized simulation results.

Funding: This work has been supported by the Cluster of Excellence “Precision Physics, Fundamental Interactions, and Structure of Matter” (PRISMA++ EXC 2118/2) funded by the German Research Foundation (DFG) within the German Excellence Strategy (Project ID 390831469).

Institutional Review Board Statement: Not applicable.

⁷ <http://axionbloch.readthedocs.io>

⁸ <http://www.sphinx-doc.org>

Informed Consent Statement: Not applicable.

Data Availability Statement: The data and code that generate this manuscript are available at <https://github.com/Yuzhe98/AxionBloch-paper>.

Acknowledgments: The author acknowledges helpful discussions with Hendrik Bekker, Dmitry Budker, and Alexander Sushkov, and thanks the CASPER Collaboration for its support. The author is grateful to the communities supporting the open-source software and tools: git, NumPy, SciPy⁹ [52], matplotlib¹⁰, astropy, pybind11, IPython¹¹ [53], Jupyter¹² [54], pytest, Sphinx, GitHub¹³ and Read the Docs¹⁴.

Conflicts of Interest: The author declares no conflicts of interest.

References

1. Peccei, R.D.; Quinn, H.R. Constraints imposed by CP conservation in the presence of pseudoparticles. *Phys. Rev. D* **1977**, *16*, 1791–1797.
2. Weinberg, S. A New Light Boson? *Phys. Rev. Lett.* **1978**, *40*, 223–226.
3. Peccei, R.D.; Quinn, H.R. CP Conservation in the Presence of Pseudoparticles. *Phys. Rev. Lett.* **1977**, *38*, 1440–1443.
4. Wilczek, F. Problem of Strong P and T Invariance in the Presence of Instantons. *Phys. Rev. Lett.* **1978**, *40*, 279–282.
5. Svrcek, P.; Witten, E. Axions in string theory. *Journal of High Energy Physics* **2006**, *2006*, 051.
6. Kim, J.E. Light pseudoscalars, particle physics and cosmology. *Physics Reports* **1987**, *150*, 1–177. [https://doi.org/10.1016/0370-1573\(87\)90017-2](https://doi.org/10.1016/0370-1573(87)90017-2).
7. Jaeckel, J.; Ringwald, A. The Low-Energy Frontier of Particle Physics. *Annual Review of Nuclear and Particle Science* **2010**, *60*, 405–437. <https://doi.org/10.1146/annurev.nucl.012809.104433>.
8. Arvanitaki, A.; Dimopoulos, S.; Dubovsky, S.; Kaloper, N.; March-Russell, J. String axiverse. *Phys. Rev. D* **2010**, *81*, 123530. <https://doi.org/10.1103/PhysRevD.81.123530>.
9. Preskill, J.; Wise, M.B.; Wilczek, F. Cosmology of the invisible axion. *Physics Letters B* **1983**, *120*, 127–132. [https://doi.org/https://doi.org/10.1016/0370-2693\(83\)90637-8](https://doi.org/https://doi.org/10.1016/0370-2693(83)90637-8).
10. Abbott, L.; Sikivie, P. A cosmological bound on the invisible axion. *Physics Letters B* **1983**, *120*, 133–136. [https://doi.org/https://doi.org/10.1016/0370-2693\(83\)90638-X](https://doi.org/https://doi.org/10.1016/0370-2693(83)90638-X).
11. Dine, M.; Fischler, W. The not-so-harmless axion. *Physics Letters B* **1983**, *120*, 137–141. [https://doi.org/https://doi.org/10.1016/0370-2693\(83\)90639-1](https://doi.org/https://doi.org/10.1016/0370-2693(83)90639-1).
12. Marsh, D.J.E. Axion Cosmology. *Physics Reports* **2016**, *643*, 1–79. <https://doi.org/10.1016/j.physrep.2016.06.005>.
13. Graham, P.W.; Rajendran, S. New observables for direct detection of axion dark matter. *Phys. Rev. D* **2013**, *88*, 035023. <https://doi.org/10.1103/PhysRevD.88.035023>.
14. Ouellet, J.L.; Salemi, C.P.; Foster, J.W.; Henning, R.; Bogorad, Z.; Conrad, J.M.; Formaggio, J.A.; Kahn, Y.; Minervini, J.; Radovinsky, A.; et al. First Results from ABRACADABRA-10 cm: A Search for Sub- μ eV Axion Dark Matter. *Phys. Rev. Lett.* **2019**, *122*, 121802. <https://doi.org/10.1103/PhysRevLett.122.121802>.
15. Du, N.; Force, N.; Khatriwada, R.; Lentz, E.; Ottens, R.; Rosenberg, L.J.; Rybka, G.; Carosi, G.; Woollett, N.; Bowring, D.; et al. Search for Invisible Axion Dark Matter with the Axion Dark Matter Experiment. *Phys. Rev. Lett.* **2018**, *120*, 151301. <https://doi.org/10.1103/PhysRevLett.120.151301>.
16. Millar, A.J.; Anlage, S.M.; Balafendiev, R.; Belov, P.; van Bibber, K.; Conrad, J.; Demarteau, M.; Droster, A.; Dunne, K.; Rosso, A.G.; et al. Searching for dark matter with plasma haloscopes. *Phys. Rev. D* **2023**, *107*, 055013. <https://doi.org/10.1103/PhysRevD.107.055013>.

⁹ <https://docs.scipy.org/>

¹⁰ <https://matplotlib.org/>

¹¹ <https://ipython.org/>

¹² <https://jupyter.org/>

¹³ <https://github.com/>

¹⁴ <https://about.readthedocs.com/>

17. Aja, B.; Cuendis, S.A.; Arregui, I.; Artal, E.; Barreiro, R.B.; Casas, F.J.; de Ory, M.C.; Díaz-Morcillo, A.; de la Fuente, L.; Gallego, J.D.; et al. The canfranc axion detection experiment (CADEx): Search for axions at 90 GHz with kinetic inductance detectors. *Journal of Cosmology and Astroparticle Physics* **2022**, *2022*, 044. 283–286
18. Ahn, S.; Kim, J.; Ivanov, B.I.; Kwon, O.; Byun, H.; van Loo, A.F.; Park, S.; Jeong, J.; Lee, S.; Kim, J.; et al. Extensive Search for Axion Dark Matter over 1 GHz with CAPP'S Main Axion Experiment. *Phys. Rev. X* **2024**, *14*, 031023. <https://doi.org/10.1103/PhysRevX.14.031023>. 287–289
19. Brouwer, L.; Chaudhuri, S.; Cho, H.M.; Corbin, J.; Craddock, W.; Dawson, C.S.; Droster, A.; Foster, J.W.; Fry, J.T.; Graham, P.W.; et al. Projected sensitivity of DMRadio-m³: A search for the QCD axion below 1 μ eV. *Phys. Rev. D* **2022**, *106*, 103008. <https://doi.org/10.1103/PhysRevD.106.103008>. 290–293
20. Alesini, D.; Babusci, D.; Beltrame, P.; Bossi, F.; Ciambone, P.; D'Elia, A.; Di Gioacchino, D.; Di Pirro, G.; Döbrich, B.; Falferi, P.; et al. The future search for low-frequency axions and new physics with the FLASH resonant cavity experiment at Frascati National Laboratories. *Physics of the Dark Universe* **2023**, *42*, 101370. 294–297
21. Zhong, L.; Al Kenany, S.; Backes, K.M.; Brubaker, B.M.; Cahn, S.B.; Carosi, G.; Gurevich, Y.V.; Kindel, W.F.; Lamoreaux, S.K.; Lehnert, K.W.; et al. Results from phase 1 of the HAYSTAC microwave cavity axion experiment. *Phys. Rev. D* **2018**, *97*, 092001. <https://doi.org/10.1103/PhysRevD.97.092001>. 298–301
22. Egge, J.; Leppla-Weber, D.; Knirck, S.; Ary dos Santos Garcia, B.; Bergermann, D.; Caldwell, A.; Dabhi, V.; Diaconu, C.; Diehl, J.; Dvali, G.; et al. First Search for Dark Photon Dark Matter with a madmax Prototype. *Phys. Rev. Lett.* **2025**, *134*, 151004. <https://doi.org/10.1103/PhysRevLett.134.151004>. 302–305
23. Quiskamp, A.P.; Flower, G.R.; Samuels, S.; McAllister, B.T.; Altin, P.; Ivanov, E.N.; Goryachev, M.; Tobar, M.E. Near-quantum-limited axion dark matter search with the ORGAN experiment around 26 μ eV. *Phys. Rev. D* **2025**, *111*, 095007. <https://doi.org/10.1103/PhysRevD.111.095007>. 306–308
24. Rettaroli, A.; Alesini, D.; Babusci, D.; Braggio, C.; Carugno, G.; D'Agostino, D.; D'Elia, A.; Di Gioacchino, D.; Di Vora, R.; Falferi, P.; et al. Search for axion dark matter with the QUAX–LNF tunable haloscope. *Phys. Rev. D* **2024**, *110*, 022008. <https://doi.org/10.1103/PhysRevD.110.022008>. 309–312
25. Gramolin, A.V.; Aybas, D.; Johnson, D.; Adam, J.; Sushkov, A.O. Search for axion-like dark matter with ferromagnets. *Nature Physics* **2021**, *17*, 79–84. <https://doi.org/10.1038/s41567-020-1006-6>. 313–315
26. Gavilan-Martin, D.; Lukasiewicz, G.; Padniuk, M.; Klinger, E.; Smolis, M.; Figueroa, N.L.; Kimball, D.F.J.; Sushkov, A.O.; Pustelny, S.; Budker, D.; et al. Searching for dark matter with a spin-based interferometer. *Nature Communications* **2025**, *16*, 4953. <https://doi.org/10.1038/s41467-025-60178-6>. 316–319
27. Bloch, I.M.; Ronen, G.; Shaham, R.; Katz, O.; Volansky, T.; Katz, O. New constraints on axion-like dark matter using a Floquet quantum detector. *Science Advances* **2022**, *8*, eabl8919, [<https://www.science.org/doi/pdf/10.1126/sciadv.abl8919>]. <https://doi.org/10.1126/sciadv.abl8919>. 320–323
28. Abel, C.; Ayres, N.J.; Ban, G.; Bison, G.; Bodek, K.; Bondar, V.; Chanele, E.; Crawford, C.B.; Daum, M.; Dechenaux, B.; et al. Search for ultralight axion dark matter in a side-band analysis of a ¹⁹⁹Hg free-spin precession signal. *SciPost Phys.* **2023**, *15*, 058. <https://doi.org/10.21468/SciPostPhys.15.2.058>. 324–327
29. Karanth, S.; Stephenson, E.J.; Chang, S.P.; Hejny, V.; Park, S.; Pretz, J.; Semertzidis, Y.K.; Wirzba, A.; Wrońska, A.; Abusaif, F.; et al. First Search for Axionlike Particles in a Storage Ring Using a Polarized Deuteron Beam. *Phys. Rev. X* **2023**, *13*, 031004. <https://doi.org/10.1103/PhysRevX.13.031004>. 328–331
30. Lee, J.; Lisanti, M.; Terrano, W.A.; Romalis, M. Laboratory Constraints on the Neutron-Spin Coupling of feV-Scale Axions. *Phys. Rev. X* **2023**, *13*, 011050. <https://doi.org/10.1103/PhysRevX.13.011050>. 332–334
31. Capozzi, F.; Dutta, B.; Gurung, G.; Jang, W.; Shoemaker, I.M.; Thompson, A.; Yu, J. New constraints on ALP couplings to electrons and photons from ArgoNeUT and the MiniBooNE beam dump. *Phys. Rev. D* **2023**, *108*, 075019. <https://doi.org/10.1103/PhysRevD.108.075019>. 335–337

32. Budker, D.; Graham, P.W.; Ledbetter, M.; Rajendran, S.; Sushkov, A.O. Proposal for a Cosmic Axion Spin Precession Experiment (CASPER). *Phys. Rev. X* **2014**, *4*, 021030. <https://doi.org/10.1103/PhysRevX.4.021030>.
33. Jackson Kimball, D.F.; Afach, S.; Aybas, D.; Blanchard, J.W.; Budker, D.; Centers, G.; Engler, M.; Figueroa, N.L.; Garcon, A.; Graham, P.W.; et al. Overview of the Cosmic Axion Spin Precession Experiment (CASPER). In *Proceedings of the Microwave Cavities and Detectors for Axion Research*; Carosi, G.; Rybka, G., Eds.; Cham, 2020; pp. 105–121.
34. Wang, T.; Kimball, D.F.J.; Sushkov, A.O.; Aybas, D.; Blanchard, J.W.; Centers, G.; O’Kelley, S.R.; Wickenbrock, A.; Fang, J.; Budker, D. Application of spin-exchange relaxation-free magnetometry to the cosmic axion spin precession experiment. *Physics of the dark universe* **2018**, *19*, 27–35.
35. Aybas, D.; Adam, J.; Blumenthal, E.; Gramolin, A.V.; Johnson, D.; Kleyheeg, A.; Afach, S.; Blanchard, J.W.; Centers, G.P.; Garcon, A.; et al. Search for Axionlike Dark Matter Using Solid-State Nuclear Magnetic Resonance. *Phys. Rev. Lett.* **2021**, *126*, 141802. <https://doi.org/10.1103/PhysRevLett.126.141802>.
36. Garcon, A.; Blanchard, J.W.; Centers, G.P.; Figueroa, N.L.; Graham, P.W.; Kimball, D.F.J.; Rajendran, S.; Sushkov, A.O.; Stadnik, Y.V.; Wickenbrock, A.; et al. Constraints on bosonic dark matter from ultralow-field nuclear magnetic resonance. *Science Advances* **2019**, *5*, eaax4539.
37. Walter, J.; Maliaka, O.; Zhang, Y.; Blanchard, J.W.; Centers, G.; Dogan, A.; Engler, M.; Figueroa, N.L.; Kim, Y.; Kimball, D.F.J.; et al. Search for axionlike dark matter using liquid-state nuclear magnetic resonance. *Phys. Rev. D* **2025**, *112*, 052008. <https://doi.org/10.1103/39nc-vr9m>.
38. Aybas, D.; Bekker, H.; Blanchard, J.W.; Budker, D.; Centers, G.P.; Figueroa, N.L.; Gramolin, A.V.; Jackson Kimball, D.F.; Wickenbrock, A.; Sushkov, A.O. Quantum sensitivity limits of nuclear magnetic resonance experiments searching for new fundamental physics. *Quantum Science and Technology* **2021**, *6*, 034007. <https://doi.org/10.1088/2058-9565/abfbbc>.
39. Dror, J.A.; Gori, S.; Leedom, J.M.; Rodd, N.L. Sensitivity of Spin-Precession Axion Experiments. *Phys. Rev. Lett.* **2023**, *130*, 181801. <https://doi.org/10.1103/PhysRevLett.130.181801>.
40. Zhang, Y.; Tümtürk, D.A.; Bekker, H.; Budker, D.; Kimball, D.F.J.; Sushkov, A.O.; Wickenbrock, A. Frequency-Scanning Considerations in Axionlike Dark Matter Spin-Precession Experiments. *Annalen der Physik* **2024**, *536*, 2300223.
41. Dror, J.A.; Murayama, H.; Rodd, N.L. Cosmic axion background. *Phys. Rev. D* **2021**, *103*, 115004. <https://doi.org/10.1103/PhysRevD.103.115004>.
42. Nitta, T.; Braine, T.; Du, N.; Guzzetti, M.; Hanretty, C.; Leum, G.; Rosenberg, L.J.; Rybka, G.; Sinnis, J.; Clarke, J.; et al. Search for a Dark-Matter-Induced Cosmic Axion Background with ADMX. *Phys. Rev. Lett.* **2023**, *131*, 101002. <https://doi.org/10.1103/PhysRevLett.131.101002>.
43. O’Hare, C.A.; Pierobon, G. Fine-grained dark matter substructure and axion haloscopes, 2025, [2509.14874].
44. Gan, X.; Kim, H.; Mitridate, A. Probing quadratically coupled ultralight dark matter with pulsar timing arrays. *Phys. Rev. D* **2026**, *113*, 063034. <https://doi.org/10.1103/15g1-gklx>.
45. Huang, X.; Ma, X.; Xu, Z.; Bloch, I.M.; Wei, K. Earth Matter Enhanced Axion Dark Matter Search, 2026, [arXiv:hep-ph/2602.20260].
46. The Astropy Collaboration.; Robitaille, Thomas P.; Tollerud, Erik J.; Greenfield, Perry.; Droettboom, Michael.; Bray, Erik.; Aldcroft, Tom.; Davis, Matt.; Ginsburg, Adam.; Price-Whelan, Adrian M.; et al. Astropy: A community Python package for astronomy. *A&A* **2013**, *558*, A33. <https://doi.org/10.1051/0004-6361/201322068>.
47. Harris, C.R.; Millman, K.J.; van der Walt, S.J.; Gommers, R.; Virtanen, P.; Cournapeau, D.; Wieser, E.; Taylor, J.; Berg, S.; Smith, N.J.; et al. Array programming with NumPy. *Nature* **2020**, *585*, 357–362. <https://doi.org/10.1038/s41586-020-2649-2>.
48. Carr, H.Y.; Purcell, E.M. Effects of Diffusion on Free Precession in Nuclear Magnetic Resonance Experiments. *Phys. Rev.* **1954**, *94*, 630–638.
49. Abragam, A.; Carr, H.Y. The Principles of Nuclear Magnetism. *Physics Today* **1961**, *14*, 56–58.
50. Evans, N.W.; O’Hare, C.A.J.; McCabe, C. Refinement of the standard halo model for dark matter searches in light of the Gaia Sausage. *Phys. Rev. D* **2019**, *99*, 023012. <https://doi.org/10.1103/PhysRevD.99.023012>.

51. Gramolin, A.V.; Wickenbrock, A.; Aybas, D.; Bekker, H.; Budker, D.; Centers, G.P.; Figueroa, N.L.; Jackson Kimball, D.F.; Sushkov, A.O. Spectral signatures of axionlike dark matter. *Phys. Rev. D* **2022**, *105*, 035029. 392
52. Virtanen, P.; Gommers, R.; Oliphant, T.E.; Haberland, M.; Reddy, T.; Cournapeau, D.; Burovski, E.; Peterson, P.; Weckesser, W.; Bright, J.; et al. SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods* **2020**, *17*, 261–272. <https://doi.org/10.1038/s41592-019-0686-2>. 393
53. Perez, F.; Granger, B.E. IPython: A System for Interactive Scientific Computing. *Computing in Science & Engineering* **2007**, *9*, 21–29. <https://doi.org/10.1109/MCSE.2007.53>. 394
54. Granger, B.E.; Pérez, F. Jupyter: Thinking and Storytelling With Code and Data. *Computing in Science & Engineering* **2021**, *23*, 7–14. <https://doi.org/10.1109/MCSE.2021.3059263>. 395

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content. 396